

DuckDB path

warehouse/air_quality.c

Result limit

10

-

+

SQL repairs

2

-

+

Examples

Which Indian city has the worst air quality?

Compare Delhi and Mumbai PM2.5 in 2020 by month.

Which cities had the highest average AQI in 2019?

Show the monthly AQI trend for Kolkata.

Which stations reported the most observations?

✓ OpenAI key

✓ Warehouse

Air Quality Analyst

Question

Compare Delhi and Mumbai PM2.5 in 2020 by month.

Run Analysis

Intent

Compare mont...

Metric

PM2.5

Chart

line

Repairs

0

Run saved to: runs/20260618_162136_compare-delhi-and-mumbai-pm2-5-in-2020

Insight

2020 Monthly PM2.5 Comparison: Delhi vs. Mumbai

In 2020, PM2.5 levels varied significantly between Delhi and Mumbai, with Delhi generally exhibiting higher average concentrations across the recorded months.

Observations

- Delhi's highest average PM2.5 in January was 157.113 µg/m³, indicating severe air quality issues.
- Mumbai's PM2.5 levels were lower overall, peaking at 72.8706 µg/m³ in January, but fell sharply to 14.6761 µg/m³ by May.
- By July, Mumbai recorded its lowest monthly average of 3.84 µg/m³, while Delhi maintained a higher level of 54.01 µg/m³.

Caveats

The data only includes selected months (January, February, March, April, May, June, and July), which may not fully represent the year.

The lack of data for certain months limits a complete annual comparison between the two cities.

PM2.5 values may fluctuate due to various external factors not accounted for, such as meteorological conditions.

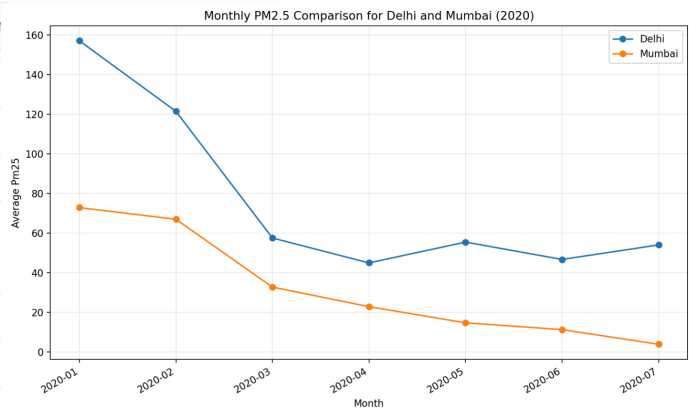
Follow-up questions

- What were the factors contributing to the fluctuations in PM2.5 levels in each city during 2020?
- Can we obtain monthly data for August through December 2020 for a more comprehensive analysis?
- How do the 2020 PM2.5 levels compare to previous years for both Delhi and Mumbai?

Result

month	City	average_pm
2020-04	Mumbai	
2020-03	Delhi	
2020-05	Mumbai	
2020-05	Delhi	
2020-03	Mumbai	
2020-06	Delhi	
2020-07	Delhi	
2020-02	Mumbai	
2020-01	Delhi	157.1126
2020-04	Delhi	44.94

Chart



Download result CSV

Generated SQL

Generated by the planner agent and validated before execution.

```
SELECT strftime('%Y-%m', "Date") AS month, "City", AVG("PM2.5") AS average_pm25
FROM city_day
WHERE "City" IN ('Delhi', 'Mumbai') AND "Date" BETWEEN '2020-01-01' AND '2020-12-31'
GROUP BY month, "City";
```

Validation

SQL is read-only and safe to run.

Result has 14 row(s) and 3 column(s).

Chart columns exist in the result.

Run log: runs/20260618_162136_compare-delhi-and-mumbai-pm2-5-in-2020

